

Grade 8
Unit 4
Lesson 9 Practice

Name _____
Date _____
Period _____

For each expression in questions 1-8, generate an equivalent expression with positive exponents by applying one or more laws of exponents.

1. 4^{-3}

2. $(d)^{-9}$

3. g^{-4}

4. $\left(\frac{1}{a}\right)^{-6}$

5. $\left(\frac{6}{t}\right)^{-7}$

6. $\frac{7k^{-8}}{k^2}$

7. $\frac{9x^9}{x^{15}}$

8. $-5y^{-5} \cdot -8y^{-1}$

Find the value of r that makes the equation true for questions 9–10.

9. $\frac{x^{13}}{x^r} = x^5$

10. $\frac{x^3}{x^r} = x^9$